

MOHD ANAS

8081233871 • anas.ahamad955@gmail.com • linkedin.com/in/anas-um • github.com/Anas-github-acc

EDUCATION

B.Tech, Information Technology

IIIT Vadodara, Gujarat, India

Expected Graduation: Jan 2027

7.43 / 10.0 GPA

Coursework: Backend System, Distributed System, Software Design, Computer Networking, Operating Systems, Database Systems, Data Structures & Algorithms, Machine Learning, Artificial Intelligence

PROFESSIONAL EXPERIENCE

Aarogya Kaya | SDE Intern | Remote

July 2025 - Jan 2026

- Built Go Backend services for Shiprocket and Instagram Graph API integrations, implementing **CQRS architecture** and a **three-tier caching strategy (runtime, Redis, and database)**, reducing average read latency from 420ms to 38ms (91%) and maintaining p95 response times below 250ms.
- Refactored 15+ Backend modules using **SOLID principles** and **Design Patterns** (such as **Dependency Injection**), reducing code duplication by 30% and improving feature delivery speed.
- Engineered data synchronization pipelines for Shiprocket and Instagram Graph API integrations, supporting 500+ daily synchronization events with 99.9% data consistency through retry mechanisms and fault-tolerant processing.

Open Source Contributor | Remote

July 2025 - Present

- Contributed a merged Pull Request to **Next.js (Vercel)**, migrating the patch-next utility from CommonJS to TypeScript ESM and reducing contributor setup overhead by eliminating repeated dependency reinstalls.
- Contributed **22 merged Pull Requests** and opened **12 Issues** across open-source projects, fixing bugs, , implementing features, and improving developer tooling.

PROJECTS

a3.store | Distributed Key-value Store | Python, EKS, AWS, SQLite, Prometheus

Live | GitHub

- Developed a **leaderless** Distributed key-value store implementing replication, **failure detection**, synchronization, gossip-based membership, and anti-entropy repair across a 3-node cluster.
- Deployed the system on **AWS EKS** using Kubernetes StatefulSets and persistent volumes, managing rolling upgrades, pod restarts, and **self-healing behavior**.
- Implemented **operational dashboards** and **disaster-recovery workflows** to make the system debuggable, reliable, and usable in production-like environments.

Shop-Intelligence | Customer Behavior Analytics using Computer Vision | Python, Next.js, Pytest

Live | GitHub

- Designed and implemented a real-time customer behavior analytics pipeline for 4+ synchronized CCTV streams, achieving 15 FPS processing using **YOLOv8**, **ByteTrack**, and ReID-based cross-camera visitor tracking.
- Engineered a concurrent video-processing architecture using **multiprocessing workers**, multi-threading, **Message Queue** to process camera streams in parallel and reduce frame-processing latency to 50ms.
- Architected a global visitor registry and **Event-Driven analytics engine** that generated real-time events for Backend retail dashboards and customer behavior analytics.

TECHNICAL SKILLS

Languages & Frameworks: Python, Go, C/C++, JavaScript, TypeScript, Rust, SQL, Node.js, React, Next.js, gRPC

Tools & Technologies: Git, GitHub, Docker, **Kubernetes**, **AWS**, EC2, EKS, IAM, VPC, Terraform, **Linux**, Kafka, Jenkins, Redis, SQLite, Prometheus, Grafana, Loki

AI & LLMs: HuggingFace, LangChain, PyTorch, NLP, Deep Learning, Generative AI

ACHIEVEMENT

- Codeforces Pupil (1245)
- Solved 500+ DSA problems across Codeforces, LeetCode, Codechef and GFG

LEADERSHIP & ACTIVITIES

Technical Committee, IIIT Vadodara - Core Member

March 2024 - Present

- Coordinated an intra-college hackathon for 150+ participants and mentored **25+ teams** for project Development.